

ch1 Temperature and Heat Transfer

Items:

Q2

Q1

*2.

Figure 2.1

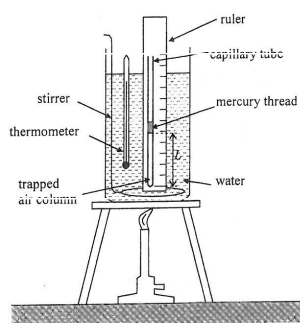


Figure 2.1 shows an air column trapped by a small mercury thread inside a uniform capillary tube. The set-up is heated by a water bath. The length of the air column L is measured at various temperature θ . Some of the results are tabulated below:

Temperature $\theta / ^\circ\text{C}$	20	92
Length of air column L / mm	64	80

- (a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped air reaches the same temperature as the water. (2 marks)

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- (b) Assume that length L increases linearly with temperature θ throughout.

- (i) Estimate the length of the air column when the temperature indicated by the thermometer is 65°C . (2 marks)

- (ii) Find the 'absolute zero' as obtained from this experiment. (2 marks)

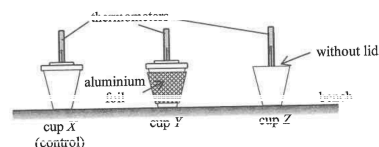
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Q1

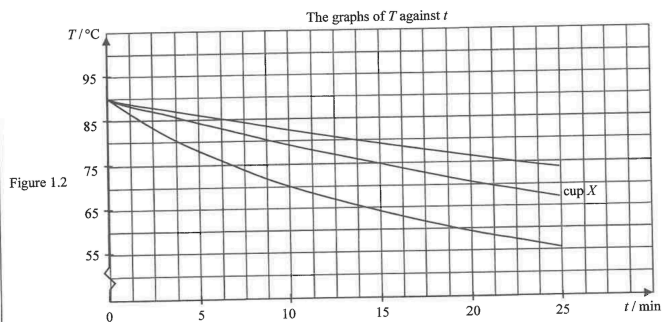
1. A student conducts an experiment to study how to best retain the warmth of water by using different types of cups. Each cup is filled with 250 cm³ of hot water and cup X serves as a control.

Figure 1.1



cup	wrapping	with a lid
X	no	yes
Y	aluminium	yes
Z	no	no

When the water temperature is 90 °C, the student starts taking thermometer readings every minute. Figure 1.2 shows the variation of temperature (T) of water in the cups with time (t) elapsed.



- (a) Suggest a reason why this experiment begins with the same initial water temperature (90 °C). (1 mark)

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- (b) Explain why all the curves become less steep as time elapses. (2 marks)

- (c) (i) On Figure 1.2, indicate the respective curves that correspond to the results of cup Y and cup Z. (1 mark)

- (ii) Explain your answer by referring respectively to the main heat transfer process. (3 marks)

- (d) Suggest a material for making cups so that heat loss via conduction can be reduced. (1 mark)